

TECTONIC SHIFT – ICTE EFFECT

TODAY CIVILIAN APPLICATION LEAD THE WAY—HEART OF THE UPGRADES OF PLATFORMS

Generation	Year of introduction	Data rates	Component technologies
1G	1981-1983	(None)	Advanced Mobile Phone System (AMPS)
First Car Phones			
			Global System for Mobile Communication (GSM)
2G	1991-1995	10 kbps	Time Division Multiple Access (TDMA)
Mass Market for Cell Phones			
			Code Division Multiple Access (CDMA)
2.5G	1999	64-144 kbps	General Packet Radio Service (GPRS)
Wireless Net Services			
			Enhanced Data Rates for Global Evolution (EDGE)
			Intermediate Specification (IS-95B)
3G	2001-2005	384-2000 kbps	Wideband CDMA (WCDMA)
Location-Based Services			
			Universal Wireless Communication (UWC-136)
			CDMA 2000
4G	2010	10-100 Mbps	Orthogonal Frequency Division Multiplexing (OFDM)
Broadcast-Quality Video			

(Courtesy: Tata Power, SED)

private players and even for the SMEs to act as prime contractors (overall responsibility), sub-contractors for design, sub-contractors for production, specific component suppliers, etc. Even the public-private partnership concept may be strengthened by sharing of prototype making and serial production activities.

In this scenario, a customised value-chain transformation is the best way to savour the changes.

Tectonic shifts in technology

No doubt, extreme dependence on technology and innovation brings continuous changes on the strategic electronics technology front. Not only the products but the testing tools are also evolving as per the stringent quality standard requirements.

Kuber sees 3G platform-based night-vision devices, non-interfering sonars and battlefield radars as the important fields requiring technological upgradation. He also puts an extra emphasis on communication technologies.

According to Ghosh, "ICTE is at the heart of network-centric operations

With network-centric operations being a force multiplier for modern warfare, India is in a unique position to leverage its world-class ICTE capabilities.

and a force multiplier."

For the next five years, non-platform and non-weapon based major ICTE programmes in the MoD pipeline will require over Rs 200 billion. These programmes will focus on development of tactical communication systems and battlefield management systems for the army. The other areas demanding technology upgradation are electronic warfare, Net-centric operations for navy, naval combat systems, air defence and operational data link upgrades, radar and sensor modernisation, naval combat systems, and artillery and tanks systems.

Return on investment

The return on investment (ROI) in strategic electronics depends on MoD's procurement and offsets. The

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